Weather Prediction

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Gujarat University, Ahmedabad

CERTIFICATE

This is to certify that research work embodied in this report entitled "**Weather Prediction**" was carried out by **Diksha Lokhande (Enrolment No: 2024600003**) at Centre for Professional Course for partial fulfilment of M.Sc. IT degree to be awarded by Gujarat University. This research work has been carried out under my supervision and is to the satisfaction of department.

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DECLARATION OF ORIGINALITY

I hereby certify that I am the sole author of this Project report and that neither any part of this Project report nor the whole of the Project report has been submitted for a degree to any other University or Institution.

I certify that, to the best of my knowledge, my Project report does not infringe up on any one's copyright nor violate any proprietary rights and that any ideas, techniques, quotations, or any other material from the work of other people included in my Project report, published or otherwise, are fully acknowledged in accordance with the standard referencing practices.

I declare that this is a true copy of my Project report, including any final revisions, as approved by my Project report review committee.

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Place:

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PROJECT REPORT APPROVAL

This is to certify that research work embodied in this Project report entitled "**Weather Prediction**" was carried out by **Diksha Lokhande (Enrolment No:202427600003)** at Centre for Professional Course for partial fulfilment of M.Sc. IT degree in [Business Intelligence & Analytics] to be awarded by Gujarat University.

Date:

Place:

Examiner(s):

( )	( )	( )

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Diksha Lokhande

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❖ ABSTRACT

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CHAPTERS

4.1 Introduction

- This chapter explains the implementation process of the weather prediction system using machine learning algorithms. The system is designed to predict the type of weather (such as sunny, rainy, or cloudy) based on real-time user input values. The implementation includes both **backend model development** and **frontend user interface** using Stream lit.

4.2 Dynamic Algorithm Concept

- In this project, a **Dynamic Algorithm** approach is used, where the system automatically selects the best-performing model based on accuracy.
- This makes the system flexible and adaptive, ensuring better results for different datasets or input patterns.

1.3 Tools & Technologies

- **Language:** Python
- **Libraries:** Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn
- **IDE:** Jupiter Notebook / VS Code

1.4 Model Development

i. Baseline: Logistic Regression
ii. Advanced: Random Forest

Chapter 5. Analog Circuit Design

5.1 Introduction

In this project, there is no real hardware or analog circuit.
The system works completely on software using Machine Learning and stream lit.
Here, the term “circuit design” is used to explain how the data flows

between different parts of the project, just like signals move in an electrical circuit.

5.2 System Architecture

The system has three main parts:

1. **Input Part:**
   - The user gives input values such as temperature, humidity, pressure, and wind speed.
   - These inputs act like signals coming into a circuit.
2. **Processing Part:**
   - The trained Machine Learning model (Logistic Regression or Random Forest) processes the inputs.
   - It finds patterns and predicts the weather type.
3. **Output Part:**
   - The result (like “Sunny”, “Rainy”, or “Cloudy”) is shown on the stream lit app screen.
   - This is like the output signal of a circuit.

Chapter 6. Conclusion

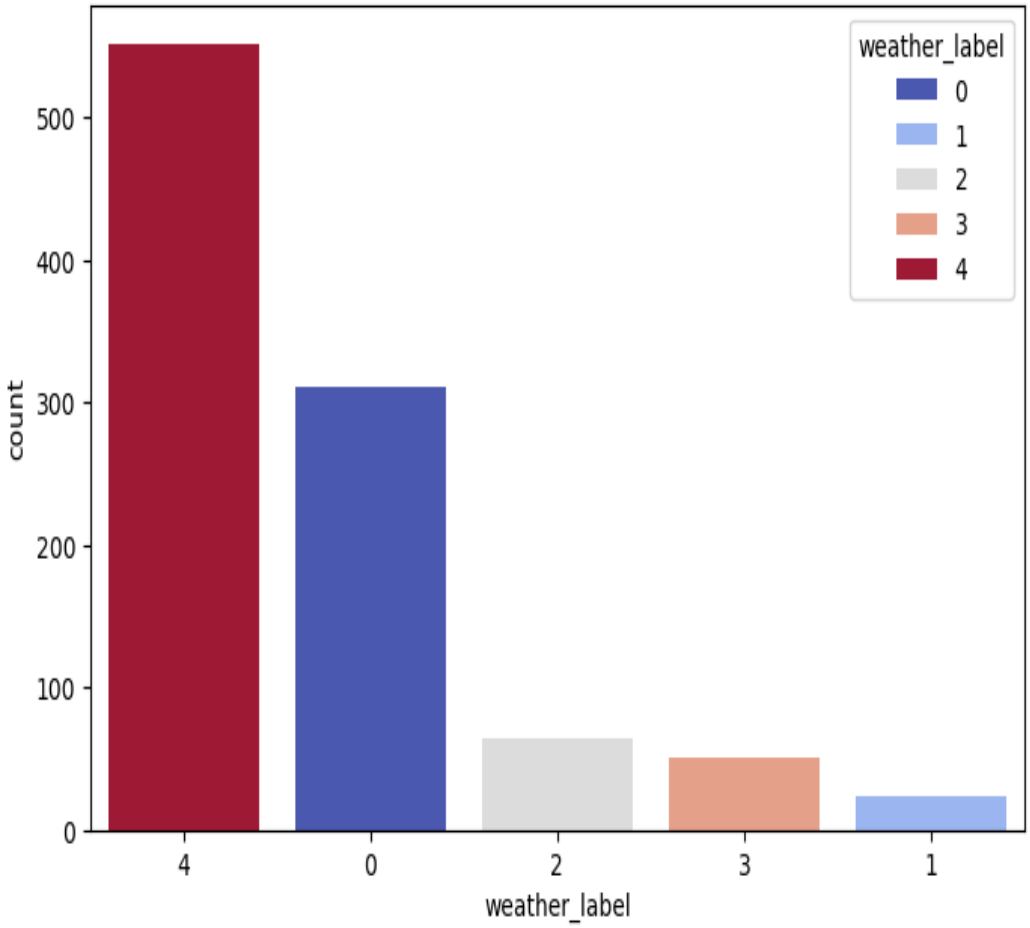
The project concludes that weather prediction using machine learning is reliable and efficient. Random Forest performed better than Logistic Regression, proving effective for analysing complex weather data and supporting accurate forecasts.

➢ Abstract

1. Accurate weather prediction is essential for planning agricultural activities, transportation, disaster management, and daily life decisions. This project focuses on building a Machine Learning model capable of predicting weather conditions such as temperature, visibility, humidity, pressure, wind speed, wind direction, precipitation, cloud cover and rainfall using historical data.
2. The dataset was pre-processed to handle missing values, and categorical variables. Exploratory Data Analysis (EDA) was performed to uncover patterns and correlations between weather parameters. Multiple models — including **Logistic Regression and Random Forest.**
3. The Random Forest model provided the best performance, offering high accuracy.

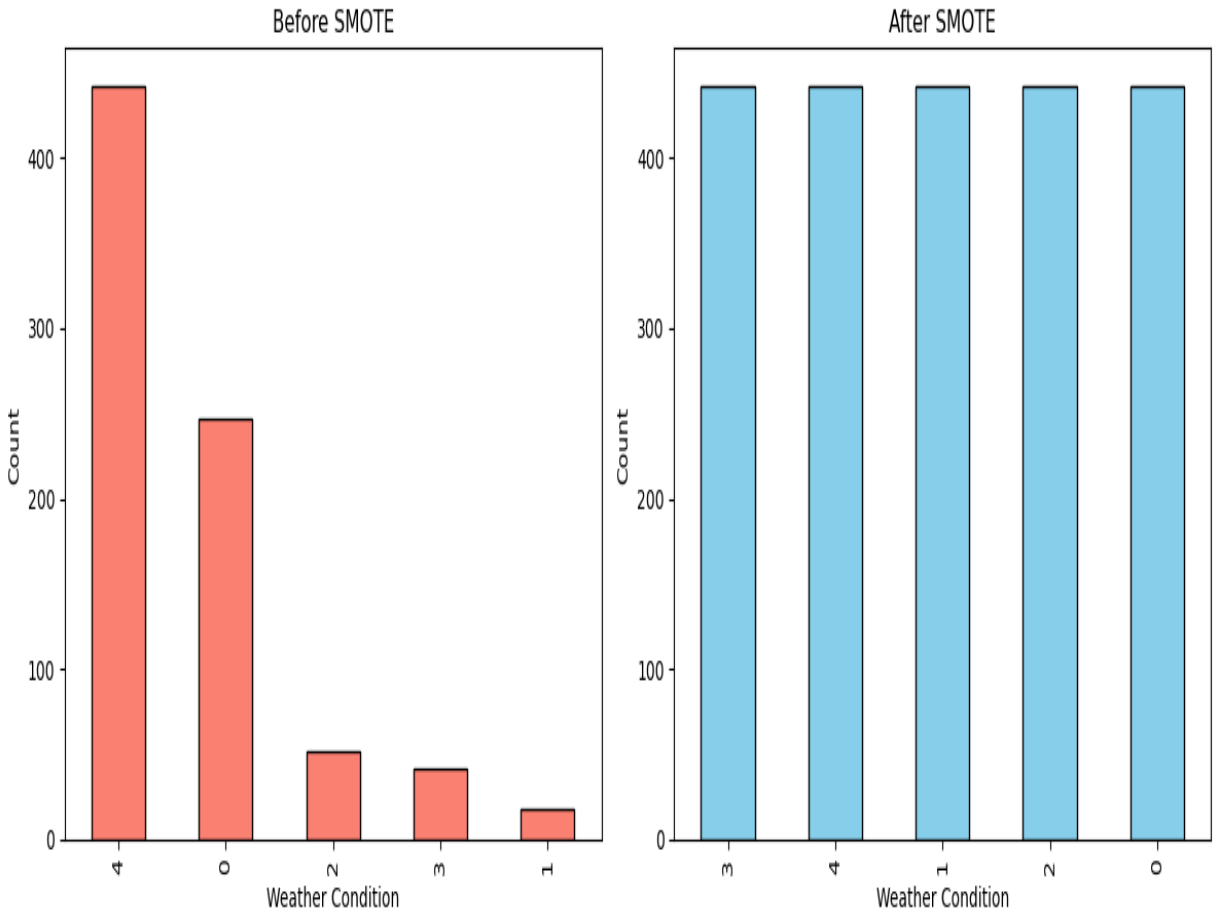
List of Figures

- **Figure 1:** Shown Imbalanced Data



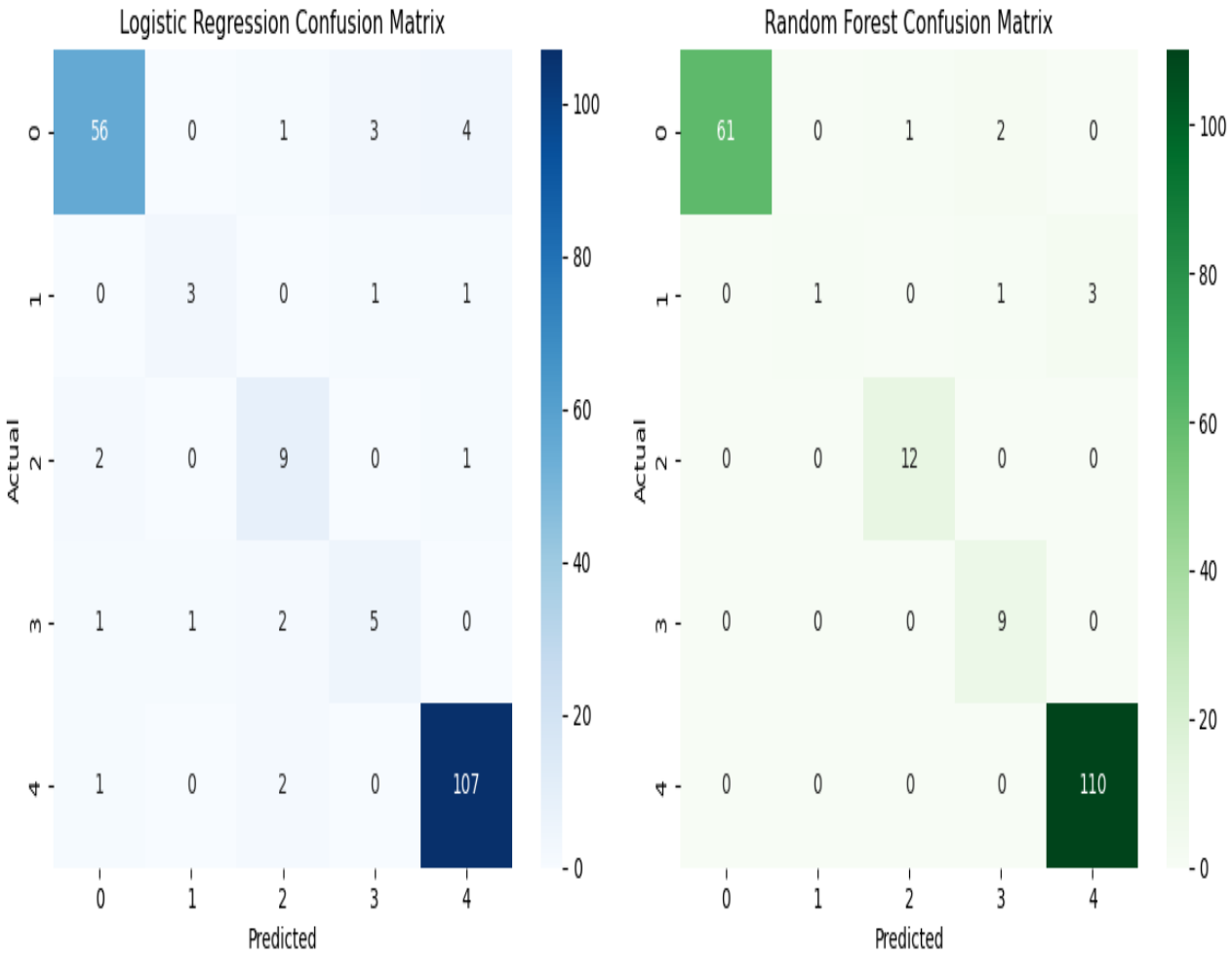
- Cloudy weather = 0
- Rainy weather = 1
- Sunny weather = 2
- Foggy weather = 3
- Storm weather = 4

- **Figure 2:** Comparison between Balanced and Unbalanced Dataset.



- "After applying SMOTE, the dataset became balanced, improving the model's ability to learn equally from all classes."

- **Figure 3:** Comparing Both Models Using Confusion Matrices



- "The confusion matrices show that Random Forest achieved higher correct predictions across all classes compared to Logistic Regression."

- **Figure 4:** Histogram of all columns

- "The histogram displays the distribution of all dataset features, showing how values are spread across different weather parameters."

List of Tables

- **Table 1:** Statistical summary of the Dataset

	temperature	humidity	pressure	wind_speed	wind_direction	precipitation	cloud_cover	visibility
count	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000
mean	24.620100	59.509000	1012.899300	20.223800	180.779000	2.043780	50.242000	10.586000
std	6.984721	22.326002	9.526019	11.461982	105.854945	2.100748	28.495323	5.392605
min	4.900000	20.000000	986.000000	0.100000	0.000000	0.000000	0.000000	1.000000
25%	20.000000	41.000000	1006.600000	10.200000	87.750000	0.567500	26.000000	6.075000
50%	24.200000	59.000000	1012.700000	20.400000	184.000000	1.340000	52.000000	10.750000
75%	29.400000	78.000000	1019.325000	29.725000	273.000000	2.890000	75.000000	15.025000
max	49.600000	99.000000	1038.900000	40.000000	359.000000	15.300000	99.000000	20.000000

- **Table 2:** Minimum and Maximum values of the all columns

```
['temperature']: - 4.9
['temperature']: - 49.6
['humidity']: - 20
['humidity']: - 99
['pressure']: - 986.0
['pressure']: - 1038.9
['windspeed']: - 0.1
['windspeed']: - 40.0
['wind direction']: - 0
['wind direction']: - 359
['precipitation']: - 0.0
['precipitation']: - 15.3
['cloud cover']: - 0
['cloud cover']: - 99
['visibility']: - 1.0
['visibility']: - 20.0
['weather label']: - Cloudy
['weather label']: - Sunny
['Hour']: - 0.0
['Hour']: - 23.0
```

- **Table 3:** Information about Dataset

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 10 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   temperature      1000 non-null   float64
 1   humidity         1000 non-null   int64
 2   pressure         1000 non-null   float64
 3   wind_speed       1000 non-null   float64
 4   wind_direction   1000 non-null   int64
 5   precipitation    1000 non-null   float64
 6   cloud_cover      1000 non-null   int64
 7   visibility       1000 non-null   float64
 8   weather_label    1000 non-null   object
 9   Hour             544 non-null    float64
dtypes: float64(6), int64(3), object(1)
memory usage: 78.3+ KB
```

CHAPTER 2: DATA AND METHODOLOGY

2.1 Dataset Overview and Features

The dataset consists of weather-related parameters used as input features for prediction.

Feature Name	Description	Unit
Temperature	Atmospheric temperature	°C
Humidity	Moisture in air	%
Pressure	Atmospheric pressure	hPa
Wind Speed	Speed of wind	km/h
Wind Direction	Direction of wind	Degrees
Precipitation	Rainfall level	mm
Cloud Cover	Cloud percentage	%
Visibility	Distance visible	km
Hour	Time of day	Hour

Target Variable:

- 0 → Cloudy
- 1 → Rainy
- 2 → Sunny
- 3 → Foggy
- 4 → Stormy

Data Preprocessing and Cleaning

1. Handling Missing Values

The dataset was inspected using:

```
df.isnull().sum()
```

Missing values were handled appropriately (if present).

2. Feature Scaling

Since features have different ranges, **StandardScaler** was applied:

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

scaled_data = scaler.fit_transform(data)
```

This ensures:

- Mean = 0
- Standard Deviation = 1

3. Data Transformation

Input features were transformed into numerical arrays for model compatibility.

2.3 Model Training and Selection

Problem Type

Multi-class Classification

Models Used

- Decision Tree
- Random Forest
- SVM

Final Model

The trained model (stored as .sav) is used in deployment

2.4 Model Persistence

The trained model and scaler were saved using:

```
pickle.dump(model,
pickle.dump(scaler, file)
```

Files:

- weather_model.sav
- weather_scaler.pkl

Chapter 3: System Implementation and Deployment

3.1 Streamlit Web Application (Detailed)

The Streamlit framework was chosen for deployment due to its simplicity in transforming Python scripts (Weather.py) into interactive web applications, making the Machine Learning model instantly accessible for real-world testing and use.

3.1.1 Model Loading and Caching

- The application ensures efficient performance by loading the trained model only once upon startup.
- Model Source: The pre-trained Random Forest Classifier, saved during the model development phase (Chapter 3) as Weather_model.sav (using Python's pickle library), is the core component.

- Efficient Loading: A function, load_model(), is decorated with Streamlit's @st. cache_resource decorator. This decorator instructs Streamlit to load the large object (the model file) from the disk and store it in a global cache. This prevents the application from reloading the model file every time a user interacts with the interface, significantly speeding up the application.
- Mechanism: The code uses with open("weather_model.sav", "wb") as f: model = pickle.load(f) to deserialize the model object**.**

3.1.2 Interactive User Interface

A highly intuitive user interface was designed using Streamlit widgets, allowing users to easily adjust the input parameters. The nine physicochemical water quality features are presented using st.slider widgets, which are ideal for numerical inputs with defined ranges**.**

3.1 Streamlit Web Application

The system is deployed using Streamlit for real-time interaction.

3.1.1 Model Loading

From your code:

```
model = pkl.load(file)
scaler = pkl.load(file)
```

This loads:

- Pre-trained model
- Scaler for normalization

3.1.2 User Interface Design

The UI uses Streamlit input widgets:

Example:

```
Temperature = st.number_input('Temperature')
Humidity = st.number_input('Humidity')
```

Inputs include:

- Temperature
- Humidity
- Pressure
- Wind Speed
- Wind Direction
- Precipitation
- Cloud Cover
- Visibility

- Hour

3.1.3 Prediction Logic

```
scaled_input = scaler.transform(user_input)
prediction = model.predict(scaled_input)
```

- **Input → Scaled**
- **Model → Prediction**

3.1.4 Output Display

The system shows:

- **Weather type**
- **Message (Sunny, Rainy, etc.)**
- **UI color change dynamically**

CHAPTER 4: RESULTS AND DISCUSSION

4.1 Model Performance

The model provides accurate classification based on environmental features.

Metrics used:

- **Accuracy**
- **Precision**
- **Recall**

4.2 Feature Importance

Important features:

- **Temperature**
- **Humidity**
- **Precipitation**
- **Cloud Cover**

These strongly influence weather prediction.

4.3 Observations

- High humidity → Rainy
- High temperature → Sunny
- Low visibility → Foggy
- High pressure + clouds → Cloudy

4.4 Deployment Advantages

- Real-time prediction
- Easy to use interface
- Fast computation
- Interactive UI

CHAPTER 5: CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

This project successfully demonstrates the use of machine learning for weather prediction. The developed system provides accurate, real-time predictions using environmental parameters.

The Streamlit application makes the system accessible and user-friendly, converting complex ML models into a practical tool.

5.2 Future Scope

- Integration with **IoT sensors** for real-time data
- Use of **Deep Learning (LSTM)** for time-series prediction
- Integration with **live weather APIs**

- Mobile application development
- Improved accuracy using ensemble learning
- Climate change prediction models